

Seeing what you eat

When should a nutrition app refuse to answer?

COMPSCI 760 · Group 1 · Research Project Update: Methodology and Results

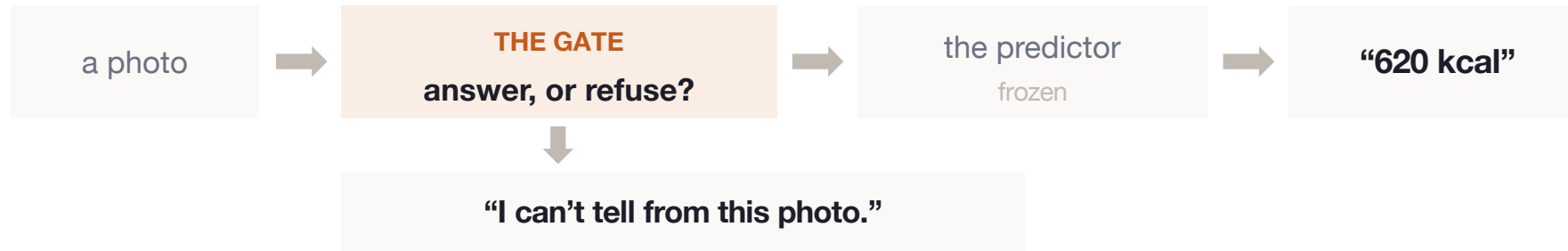
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Code and results: github.com/sushantkrishnan/seeing-what-you-eat

RECAP

The predictor is frozen. We study the gate.



THE PROBLEM

Calorie apps answer every photo with one number and never decline. A confident wrong number is worse than no number.

WHAT WE PROPOSED

Freeze the predictor. Benchmark the signals an app could use to decide whether to answer, between refusing at random and a perfect refuser.

OBJECTIVES · STATUS

O1 Benchmark

Which signals predict when the answer is wrong?

due 11 Sep · **done**

O2 Ceiling

How much is on the table, and on which axis?

due 20 Sep · **done**

O3 Ranges

Does the best signal give ranges worth having?

due 11 Oct · **first cut done**

Headline question: which signals let an app know it is guessing, and can any of them run on the phone?

SINCE THE PROPOSAL

What the markers asked, and what we changed.

FEEDBACK WE RECEIVED, AND WHAT WE DID WITH IT

Freezing the predictor and studying the gate is the right design.

Kept. Nothing on the predictor side changed.

Which evaluation metrics, exactly?

One fixed set, stated on slide 5: error at a fixed answer rate, the area under the whole curve, the share of the random-to-perfect gap captured, and what each gate refuses. Ranges get coverage, width and a proper interval score.

How will you show a difference is significant?

Every number now carries a 95% interval from a bootstrap over plate sessions, and every gate is tested against refusing at random with a Holm correction. Two proposal claims did not survive this; slide 8 says which.

CHANGES WE MADE, AND WHY

Calibration is now carved by plate session, not by dish.

98% of calibration dishes shared a scanning session with a fit dish. Fixing it moved the predictor from 73.4 to 70.9 kcal, inside its own interval.

We report the whole curve and quote the 90% answer rate.

That is where an app would sit. The proposal quoted 50%, where the picture is rosier.

The phone-camera rung is in. The hidden-food rung is dropped.

Phone: late by three weeks. Hidden food: damage is 9% of the error, and we cannot label occlusion honestly on this data.

Confidence ranges started three weeks early.

The first cut is on slide 11. It found the next problem: the largest meals are under-covered.

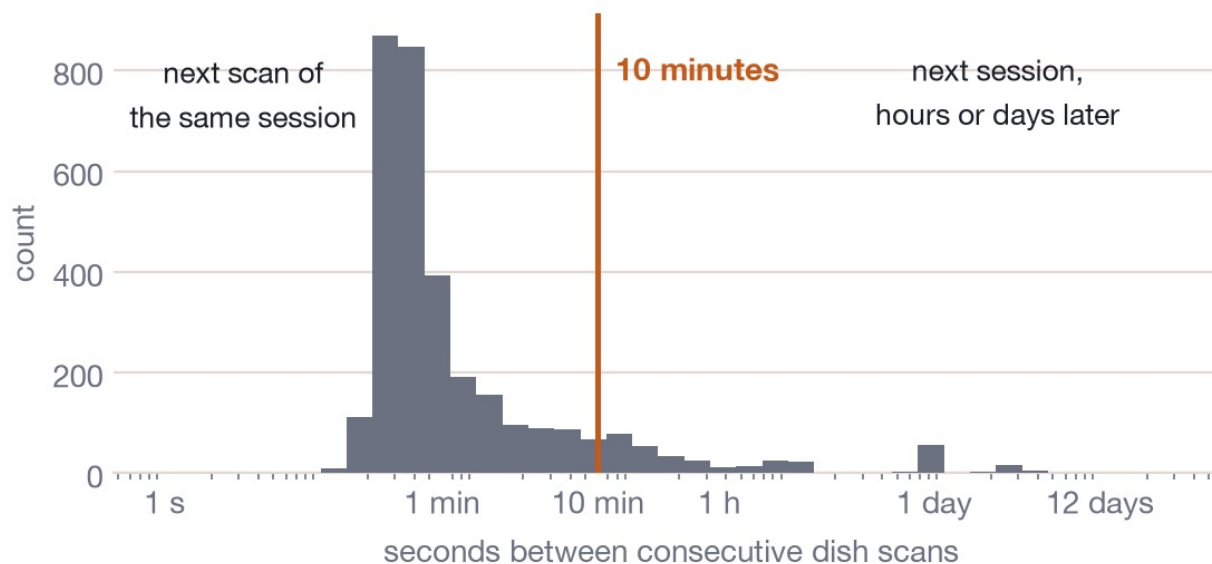
DATASET

Nutrition5k, and the unit of analysis we had missed.



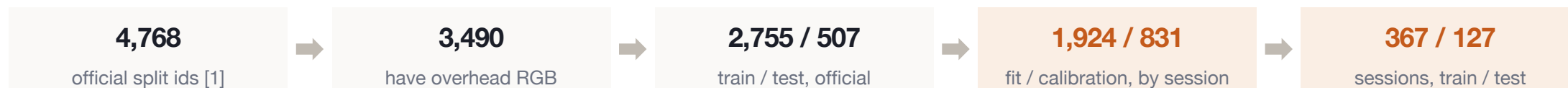
Overhead RGB, one fixed rig, every ingredient weighed [1]. Chosen because the per-ingredient truth lets us build a perfect refuser to measure against.

SCANS OF ONE PLATE ARRIVE 40 SECONDS APART



So the 507 test dishes are 127 plate sessions, and error is correlated inside a session (intraclass correlation 0.34). Sessions are the unit for the calibration curve and for every confidence interval in this talk.

SPLITS



One pass, one file, one fixed set of measures.

THE PIPELINE

Frozen backbone (CLIP ViT-B/16; SigLIP 2 and DINOv3 as checks) → 2,304-d features, cached once.

Standardise on the fit split, then a closed-form ridge head. No training loop, so no convergence curve to show.

Ridge alpha chosen on calibration over a 25-point quarter-decade grid (picked 5623). PCA $k = 64$, $B = 32$ heads, $K = 10$ neighbours: fixed before looking at test.

8 damage levels × 507 dishes × 3 backbones = 12,168 rows, every gate score in one CSV. Every result is a lookup on that file.

WHAT A GATE IS SCORED ON

Error (MAE, kcal) at a fixed answer rate; we quote 90%, and show the whole curve from 100% to 10%.

Area under that curve (AURC), and the share of the random-to-perfect gap the gate captures.

Rank correlation (Spearman ρ) between the signal and $|\text{error}|$.

What it refuses: mean size of the meals kept, and the share of 400+ kcal meals refused, against the perfect refuser's.

HOW WE TEST SIGNIFICANCE

Bootstrap over the 127 plate sessions, 2,000 resamples, recomputing every curve on the same resample: intervals and paired tests.

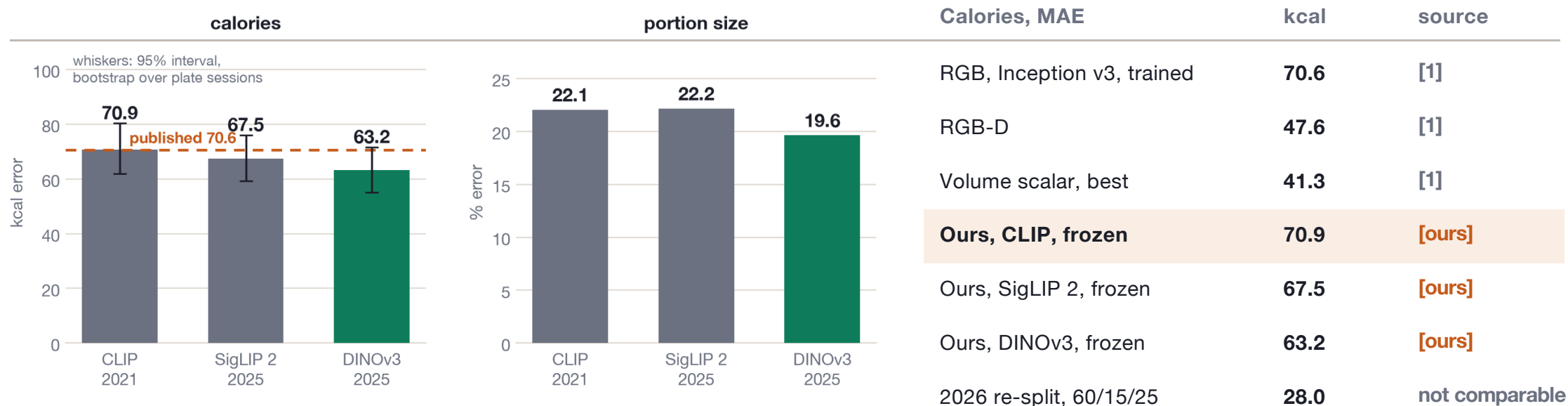
Each gate vs refusing at random at each answer rate; Holm correction across the ten candidates.

Ranges: coverage with an interval, width, share within ± 100 kcal, coverage by difficulty quartile, Winkler interval score.

Compute: one M-series laptop, ~68 images/s; the whole pipeline reruns in under 15 minutes.

Floor and ceiling on every chart: refusing at random, and a refuser that knows the true error. A gate is judged by where it sits between them, and by what it refuses.

The published RGB baseline, reproduced, no training.



507 test dishes, official split. CLIP 70.9 kcal [62, 80]: the published 70.6 sits inside the interval. Portion mass 43.8 g (22%). DINOv3 is the only backbone that moves mass.

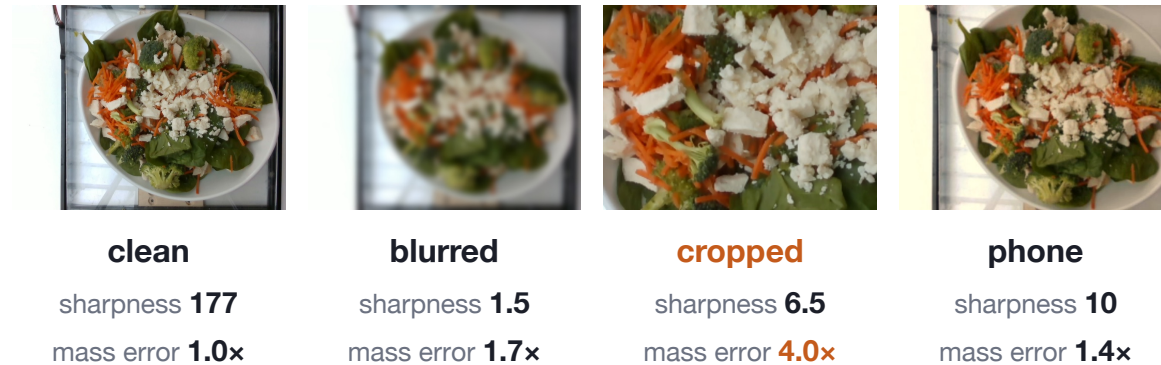
The 28.0 kcal paper re-splits Nutrition5k at random, so near-duplicate scans land on both sides. We keep [1]'s split and do not compare to it.

WHY WE DO NOT CHASE THE NUMBER a two-stage model that named ingredients and weighed them, with the mass error we actually measure (22%) and one name in five wrong, lands at 87 kcal [ours, oracle budget]. Worse than the one-step model we already have.

RESULT 1, COMPLETED

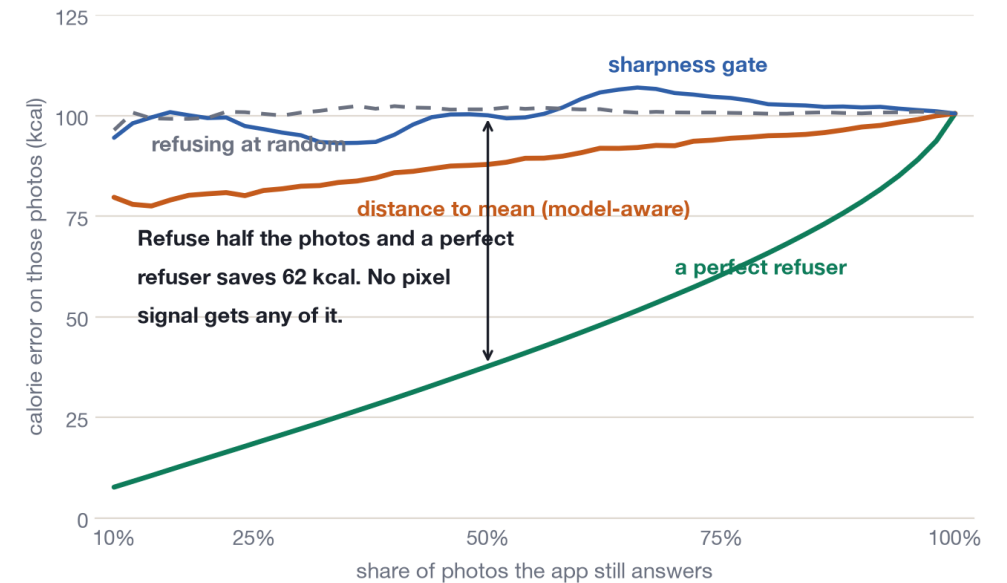
Pixel signals do not rank the error. The dish does.

ONE DISH, FOUR PHOTOGRAPHS [OURS]



Sharpness scores the cropped photo four times better than the blurred one; it carries far more error, because the plate rim is the only clue to portion size. The phone rung costs 1.2x on calories and 1.4x on mass.

REFUSE THE WORST PHOTOS, ALL 8 DAMAGE LEVELS POOLED [OURS]

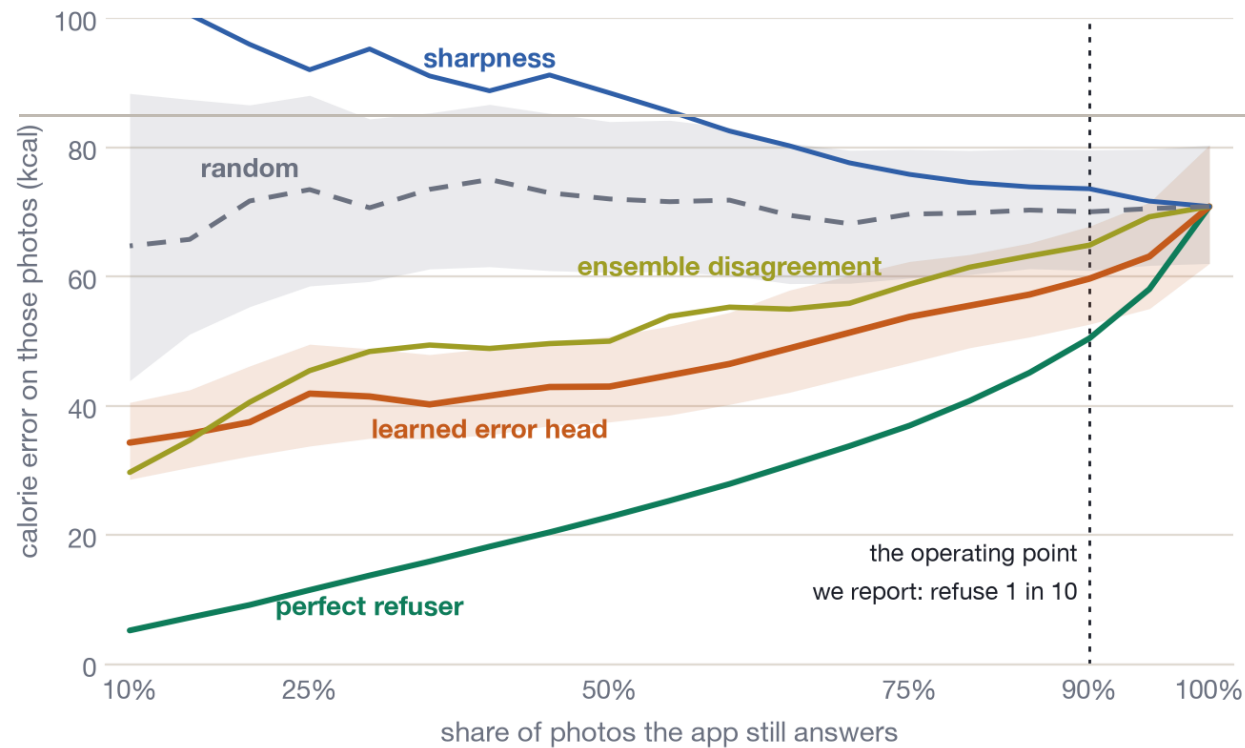


THE DAMAGE IS DETECTED, AND DETECTING IT DOES NOT HELP sharpness separates damaged photos from clean ones at $AUC \approx 1.00$. Yet over 4,056 dish × damage rows, no pixel metric ranks the error above chance ($|p| \leq 0.17$). Variance of the error: which dish 62%, damage level 9%, their interaction 29%.

A gate has to read the dish, so the useful signals come from the model, not the pixels.

RESULT 2, COMPLETED

Model-aware signals bend the curve, with intervals.



Clean test photos, CLIP. Bands are 95% intervals over plate sessions.

AT 90% ANSWERED, CLIP [OURS]

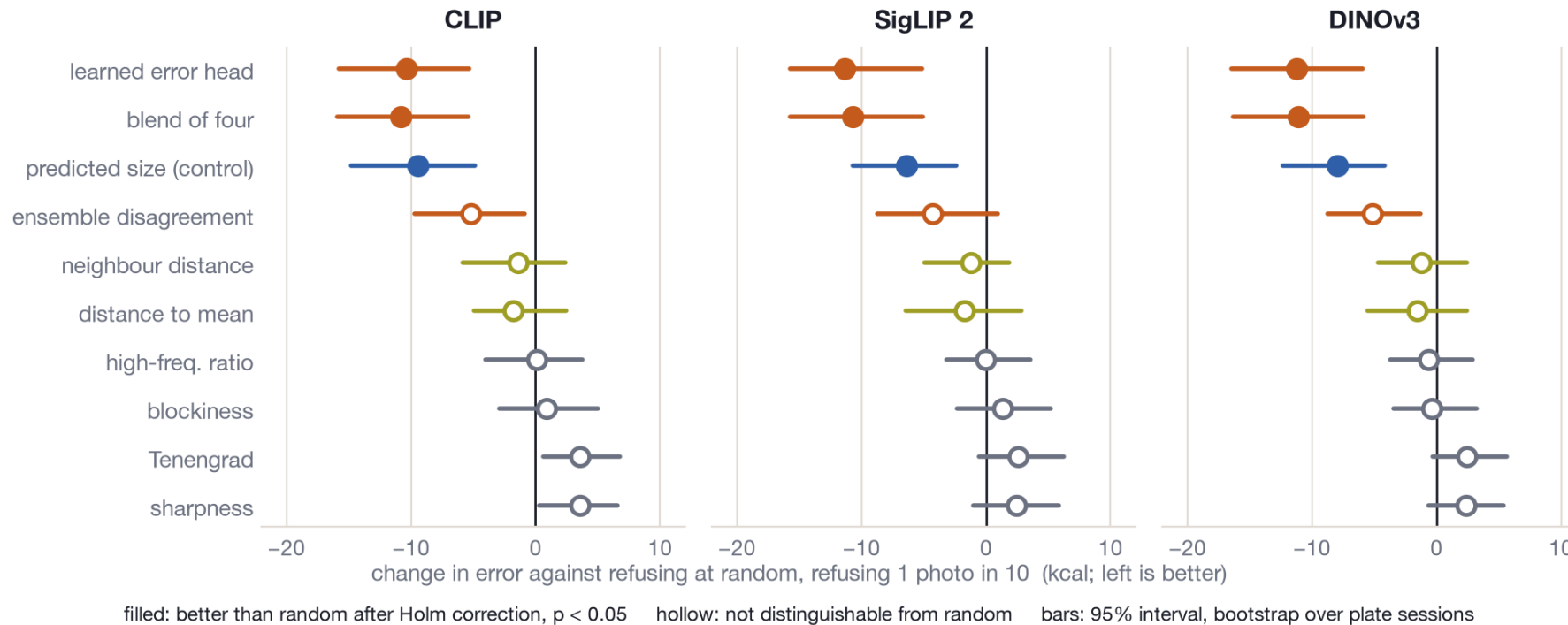
gate	kcal	vs random
perfect refuser	50.4	—
learned error head	59.7	-10.3 [-16, -5] ✓
blend of four	59.2	-10.8 [-16, -5] ✓
predicted size (control)	60.6	-9.4 [-15, -5] ✓
ensemble disagreement	64.9	-5.2 [-10, -1]
distance to mean	68.2	-1.8 [-5, 2]
neighbour distance	68.6	-1.4 [-6, 2]
random	70.0	—
sharpness	73.6	+3.6 [0, 7]

✓ = better than random after Holm correction, $p < 0.05$. Interval is the paired difference.

WHAT SURVIVED THE TEST the learned error head beats random by 10 kcal at 90% answered ($p = 0.005$). The proposal's favourite, ensemble disagreement, does not clear the correction there ($p = 0.11$) and is 5 kcal behind the error head (paired $p = 0.004$). Neither distance gate is distinguishable from random on clean photos.

RESULT 2, COMPLETED

The same answer on three backbones.



WHAT IS SIGNIFICANT

The learned error head, the blend and the control beat random on all three backbones (Holm $p < 0.01$). The head and the blend are identical: blending adds nothing.

WHAT IS NOT

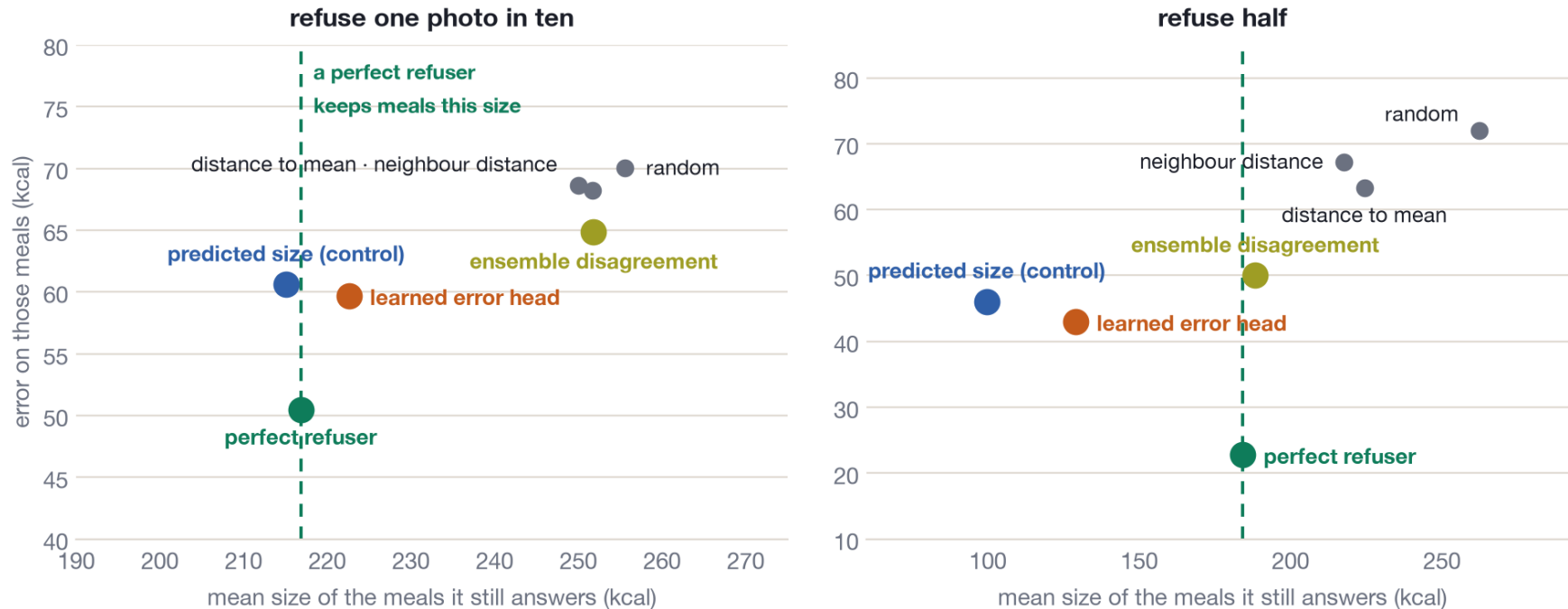
Ensemble disagreement clears the test at 50% answered, not at 90%. The two distance gates never do. No pixel metric ever does; sharpness is worse than random at 50%.

SHARE OF THE GAP CAPTURED

Learned error head 54% [41, 64], ensemble 42%, distance to mean 8% with an interval spanning zero.

RESULT 3, COMPLETED

Check what each gate refuses before believing it.



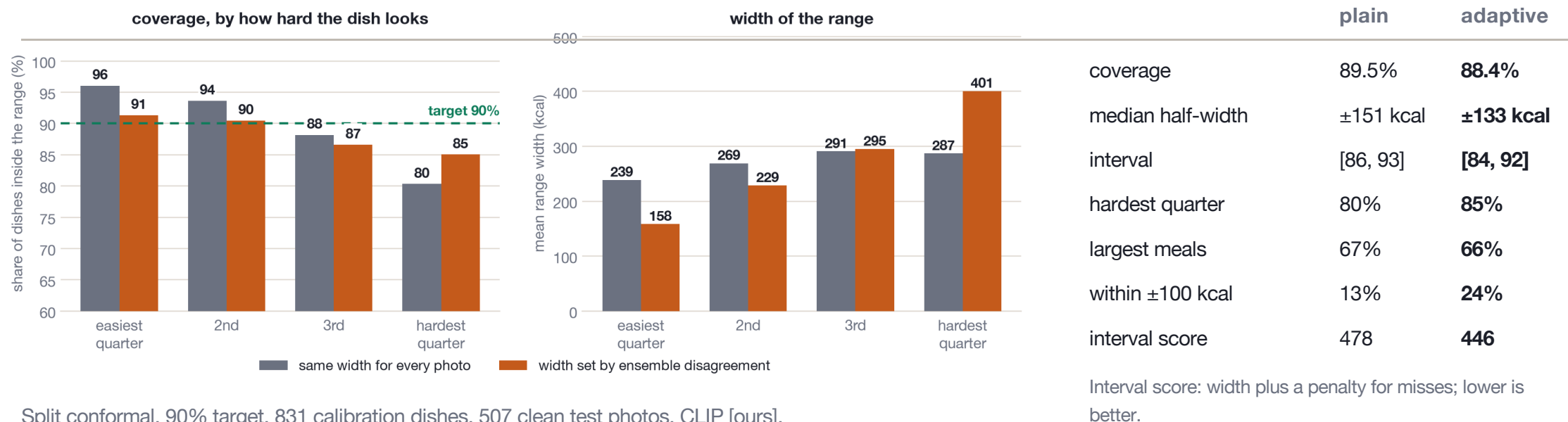
THE CONTROL error grows with portion size, so refusing big meals lowers the error without reading the photo. Scoring risk by the predicted calorie count alone does exactly that.

At 90% answered every learned gate keeps meals the size a perfect refuser keeps (error head 223 kcal, perfect 217). At 50% the control keeps 100-kcal meals and the error head 129, against 184 for a perfect refuser. Only ensemble disagreement stays with the oracle (188).

So the honest gate depends on the answer rate. We report the kept-set size beside every error, at every operating point.

RESULT 4, FIRST CUT

Ranges: honest on average, not on the hard dishes.



THREE THINGS THIS TELLS US (1) Marginal coverage hides a lot: the plain range reads 90% overall and covers the hardest quarter of dishes 80% of the time. Letting the gate signal set the width closes most of that gap on all three backbones and improves the proper score.

(2) Neither range covers the largest quarter of meals at 90% (about 67%). That is the problem O3 has to solve next, and it is the same big-meal effect as slide 10.

(3) Exchangeability is real: calibrate on clean photos and deploy on a damaged mixture and coverage drops to 81%; calibrating on the same mixture restores 90% at 28% more width. A check, not a finding.

Two milestones slipped, one moved early.

CHALLENGES AND WHAT WE DID

Correlated test dishes

Errors cluster by plate session (ICC 0.34). Unit of resampling is now the session; the calibration curve respects it.

A gate that cheats

Refusing big meals looks like skill. A control gate exposed it; kept-set size is reported at every operating point.

Rank-deficient covariance

2,304 dims on 1,924 dishes. Mahalanobis on the leading 64 components, fixed in advance.

No phone photos in the data

The phone rung is an imitation of four mild effects, and is labelled as one.

PROPOSAL TIMELINE VS ACTUAL

planned	milestone	status
17 Aug	Pipeline, 3 backbones, first results	done
25 Aug	Phone rung; hidden-food rung	late · dropped
11 Sep	All gates scored, one results file	done 17 Sep
20 Sep	Literature review	in progress
11 Oct	Confidence ranges	first cut 17 Sep
25 Oct	Final report	next

Next: size-aware ranges for large meals (CQR); the deployment mixture on all backbones; the cost tier (test-time augmentation, a second backbone); report.

WHO OWNS WHAT

Nisarg

pipeline, results file, framing

Shreya

damage levels, pixel gates

Keanu

model-aware gates, curves

Sushant

literature, writing, data

Kevin

confidence ranges

CONCLUSIONS

What we know now, and did not at the proposal.

- 1 Photo quality is not the axis.**
Damage is detected almost perfectly and explains 9% of the error. The dish explains 62%. Cheap pixel gates never beat random, on any backbone, at any answer rate.
- 2 Model-aware gates work, and we can now say how sure we are.**
At 90% answered the learned error head cuts the error by 10 kcal against random (Holm $p = 0.005$) on three backbones, keeps normal-sized meals, and captures about half the gap to a perfect refuser.
- 3 Which gate is honest depends on the answer rate.**
At 50% the same gate drifts to small meals and ensemble disagreement is the one that tracks the oracle. A control gate and the kept-set size are part of every result.
- 4 Ranges are honest on average and not yet on hard or large dishes.**
Adaptive width fixes most of the difficulty gap. The largest meals are still under-covered. That is O3's job by 11 October.

Headline question, answered so far: an app can know when it is guessing, from a signal that costs no extra inference, at the cost of refusing one photo in ten.

CLOSING

Sources, AI use, and where the code is.

- [1] Q. Thames, A. Karpur, W. Norris, F. Xia, L. Panait, T. Weyand, and J. Sim, “Nutrition5k: Towards automatic nutritional understanding of generic food,” in Proc. IEEE/CVF CVPR, 2021, pp. 8903–8911.
 - [2] B. Coburn, J. He, M. E. Rollo, S. S. Dhaliwal, D. A. Kerr, and F. Zhu, “Comprehensive evaluation of large multimodal models for nutrition analysis: A new benchmark enriched with contextual metadata,” in Proc. IEEE Int. Conf. Biomed. Health Informat. (BHI), 2025.
 - [3] P. Grother and E. Tabassi, “Performance of biometric quality measures,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 29, no. 4, pp. 531–543, Apr. 2007.
 - [4] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in Proc. ICLR, 2019.
 - [5] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
 - [6] Y. Geifman and R. El-Yaniv, “SelectiveNet: A deep neural network with an integrated reject option,” in Proc. ICML, PMLR 97, 2019.
- Re-split comparison on slide 6: V. Awasthi et al., Int. J. Intell. Eng. Syst., vol. 19, no. 2, 2026, doi:10.22266/ijies2026.0228.08.
- All figures and tables are ours unless tagged [1]. Images on slides 4 and 7 are Nutrition5k dishes [1], reproduced under the dataset licence.

GENERATIVE AI USE Claude Code (Anthropic) was used to scaffold pipeline code, generate the slides from the results files, and improve wording. The experimental design, analysis choices, results and conclusions are the group’s own and reproduce from the repository.

CODE AND RESULTS github.com/sushantkrishnan/seeing-what-you-eat · every number in this deck is regenerated from results/*.json by the deck builder